Our model $A(\hat{n})$ = the Fourier fit to Migkas et al. (2020)

Sources on the vision shell, $d = r$; $\mu = \cos b_0 \cos(\ell - \ell_0)$; $\kappa = \dfrac{GM(<D)}{H_0^2 D^3}$; $W_m = \dfrac{\sin(m\pi/4)}{m\pi/4}$ (90° sectors).

$$\underbrace{N\Big[1 + \alpha\kappa\,(3\mu^2 - 1) + \alpha\kappa\,\frac{r}{D}\Big(\tfrac{9}{2}\mu - \tfrac{15}{2}\mu^3\Big)\Big]}_{\text{our model}} = \underbrace{a_0 + \sum_{m=1}^{3}\big[c_m \cos m\ell + s_m \sin m\ell\big]}_{\text{Fourier fit}}$$

$A_{\rm fit}(\ell) = 0.9965 + 0.0296\cos\ell + 0.0434\sin\ell + 0.0636\cos 2\ell + 0.0366\sin 2\ell - 0.0299\cos 3\ell - 0.0114\sin 3\ell, \quad R^2 = 0.915$

Term by term (fit projected on $\cos m(\ell - \ell_0)$, $\ell_0 = 15°$, $c \equiv \cos b_0$):

$$\begin{aligned}
m = 0: &\quad 0.9965 = N\big[1 + \alpha\kappa(\tfrac{3}{2}c^2 - 1)\big] \\
m = 1: &\quad +0.0398 = N\alpha\kappa\,\tfrac{r}{D}\,c\Big(\tfrac{9}{2} - \tfrac{45}{8}c^2\Big)W_1 \\
m = 2: &\quad +0.0733 = N\alpha\kappa\,\tfrac{3}{2}c^2\,W_2 \\
m = 3: &\quad -0.0292 = -N\alpha\kappa\,\tfrac{r}{D}\,\tfrac{15}{8}c^3\,W_3
\end{aligned}$$

Solution: $c^2 = 0.695$ ($|b_0| = 33.5°$), $\quad \dfrac{r}{D} = 0.812 \;\Rightarrow\; \dfrac{D}{r} = 1.23$, $D \approx 17.6$ Gpc, $\quad N = 0.992$, $\quad \alpha\kappa = 0.111$.

Not reproducible by an axisymmetric model (the $\sin m(\ell - 15°)$ parts): m_1: $+0.034$, m_2: -0.000, m_3: $+0.013$.

The analytic form is a small-r/D expansion evaluated at $r/D \approx 0.8$. The full numeric model gives $R^2 = 0.776$ at $D/r \approx 1.32$.